

Please read the following carefully. *Lees die volgende noukeurig*

1. Answer all the questions in the space provided on the question paper. *Antwoord alle vrae in die spasie gelaat op die vraestel.*
2. No pencil work will be marked. Do not use "tippex". *Geen potlood-werk sal gemerk word nie. Moet nie uitwis-vloeistof ("tippex") gebruik nie.*
3. If you need more space, please use the back of the pages, and clearly indicate where the question continues. *As u meer spasie benodig gebruik die agterkant van die bladsye. Dui duidelik aan waar die vraag aangaan.*
4. Where applicable you need to draw the necessary diagrams (e.g. free-body diagrams). *Teken diagramme waar nodig (bv. vryliggaams-diagramme).*
5. Your answer needs to be written in a systematic and logical fashion. You must show that you understand and can use the relevant physics. **Only substitute numerical values in the final step of calculations! A correct answer does not imply that you will receive full marks.** *U antwoorde moet sistematies en logies neergeskryf word. U moet bewys dat u die nodige fisika verstaan en kan toepas. Numeriese waardes mag slegs in die laaste stap van 'n berekening vervang word! 'n Korrekte antwoord beteken nie dat u noodwendig vol punte sal kry nie.*
6. Units must be used when numbers are substituted else marks will be subtracted. *Eenhede moet gebruik word wanneer numeriese waardes vervang word, andersins sal punte afgetrek word.*
7. The use of the cheat sheet is only allowed subject to the rules stated on it. *Die formuleblad mag slegs gebruik word onderhewig aan die reëls daarop gedruk.*
8. All exam regulations of the University of Pretoria apply, and any contravention of it will be treated in a serious manner. *Alle eksamen regulasies van die Universiteit van Pretoria geld en enige oortreding daarvan sal in 'n ernstige lig beskou word.*

Constants and formulas / Konstantes en formules

$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$; $G = 6.674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$; $e = 1.602 \times 10^{-19} \text{ C}$; $k = 1.381 \times 10^{-23} \text{ J/K}$; $m_e = 9.109 \times 10^{-31} \text{ kg}$; $m_p = 1.673 \times 10^{-27} \text{ kg}$; $R = 8.314 \text{ J/(mol}\cdot\text{K)}$; $N_A = 6.02 \times 10^{23} / \text{mol}$; $1 \text{ atm} = 101.3 \text{ kPa}$; $1 \text{ u} = 1.6605677 \times 10^{-27} \text{ kg}$; $c = 3 \times 10^8 \text{ m/s}$; $V_{\text{sphere/sfeer}} = \frac{4}{3}\pi r^3$; $F_{\text{gravity}} = \frac{Gm_1m_2}{r^2}$; $1 \text{ litre} = 1000 \text{ cm}^3$; $M_{\text{Earth}} = M_{\text{Aarde}} = 5.97 \times 10^{24} \text{ kg}$; $\rho(\text{water}) = 1 \text{ g/cm}^3$; $r_{\text{Earth}} = r_{\text{Aarde}} = 6.37 \times 10^6 \text{ m}$

1 Question 1 / Vraag 1

[7]

Andy Warhol was a famous artist. He once said that "everyone will be world-famous for 15 minutes". *Andy Warhol was 'n befaamde kunstenaar. Hy het gesê "elke persoon sal vir 15 minute wêreld-bekend wees".*

- (a) If we define a new unit of 1 warhol = 15 minutes, how long is 1 megawarhol in years? Show the conversion factors in your calculation.

As ons definieer dat 1 warhol = 15 minute, hoe lank is 1 megawarhol in jare? Toon die omskakelingsfaktore in jou berekening.

(4)

$$\begin{aligned}
 1 \text{ megawarhol} &= \frac{1 \times 10^6 \text{ warhols}}{15 \text{ minutes}} \times \frac{60 \text{ min}}{1 \text{ hour}} \times \frac{24 \text{ hours}}{1 \text{ day}} \times \frac{365 \text{ days}}{1 \text{ year}} \\
 &= 28,56 \text{ years}
 \end{aligned}$$

- (b) Photons move at a constant speed of 300 000 km/s. How far (in meters) would a photon move in 1 Megawarhol? Show the conversion factors in your calculation. Use the given data in your calculation.

Fotone beweeg teen 'n konstante spoed van 300 000 km/s. Hoe ver (in meters) sal 'n foton beweeg in 1 Megawarhol? Toon die omskakelingsfaktore in jou berekening. Gebruik die gegewe data in jou berekening. (3)

$$1 \text{ Megawarhol} = 15 \times 10^6 \text{ minutes}$$

$$\begin{aligned} \text{Distance} &= \frac{300\,000 \text{ km}}{1 \text{ seconds}} \times \frac{60 \text{ seconds}}{1 \text{ minute}} \times \frac{15 \times 10^6 \text{ minutes}}{1 \text{ megawarhol}} \\ &= 2,7 \times 10^{14} \text{ km} / \text{megawarhol} \end{aligned}$$

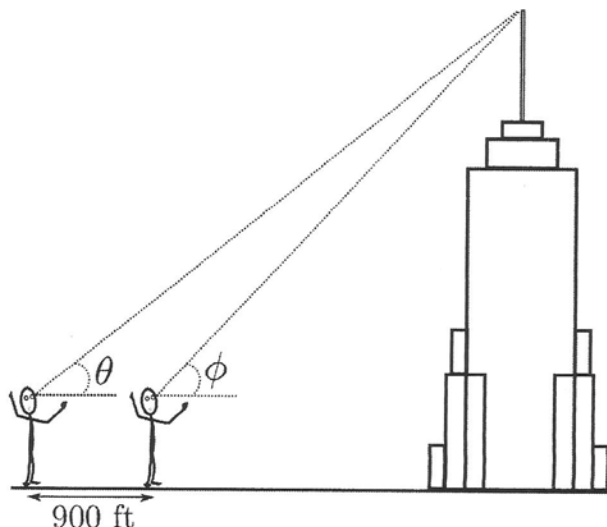
$\therefore$ The photon would move ~~$2,7 \times 10^{14} \text{ km}$~~ $2,7 \times 10^{17} \text{ m}$ in 1 Megawarhol.

2 Question 2 / Vraag 2

[6]

During a visit to New York, a tourist decides to estimate the height of the Empire State building. She measures the angle θ of elevation of the spire atop the building as 20° . After walking 900 feet (ft) closer to the building, she measures a new angle of elevation as $\phi = 25^\circ$. Use this data to **estimate** the height of the Empire State building. You can express your answer in feet. Use an annotated (labeled) sketch to explain your calculation. The sketch below is not to scale.

Gedurende 'n besoek aan New York besluit 'n toeris om die hoogte van die Empire State-gebou te bepaal. Sy meet die hoogtehoek θ van die toring bo-op die gebou as 20° . Nadat sy 900 voet (ft) nader aan die gebou geloop het, meet sy 'n nuwe hoogtehoek van $\phi = 25^\circ$. Gebruik dié data om die hoogte van die Empire State-gebou **benaderd te bepaal**. Jy kan jou antwoord in voet laat. Gebruik 'n skets met aantekeninge om jou berekening te verduidelik. Die figuur hieronder is nie volgens skaal geteken nie.



check

$$\tan \phi = \frac{y}{x-900}$$

$$\tan 25^\circ = \frac{y}{x-900}$$

$$(x-900) = \frac{y}{\tan 25^\circ}$$

$$x = \frac{y}{\tan 25^\circ} + 900 \quad (1)$$

$$\tan \theta = \frac{y}{x}$$

$$\tan 20^\circ = \frac{y}{x}$$

$$x = \frac{y}{\tan 20^\circ} \quad (2)$$

$$\frac{y}{\tan 25^\circ} + 900 = \frac{y}{\tan 20^\circ}$$

$$y + 900 = \frac{y}{\tan 20^\circ} (\tan 25^\circ)$$

$$y = \frac{y \tan 25^\circ}{\tan 20^\circ} - 900$$

$$y = 1,28(y) - 900$$

$$-0,28y = -900$$

$$y \approx 3214,29 \text{ feet} + 5 \text{ feet}$$

The height is approximately ~~3214,29 feet~~.

3219,29 feet, an extra 5 feet were added to

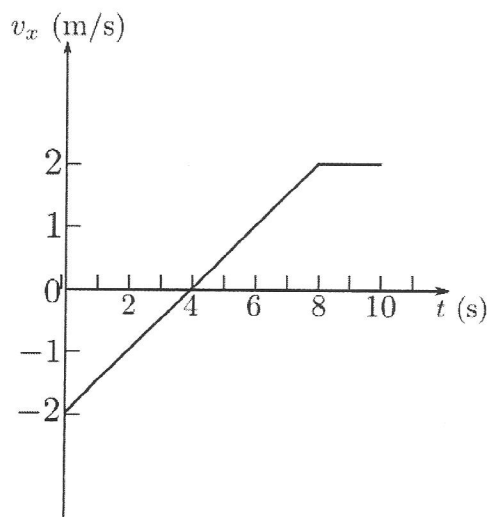
accommodate the height of the woman, as the angles provided are from eye level. An average person is about 5 feet tall.

3 Question 3 / Vraag 3

[8]

The figure shows the velocity graph of a train that starts from the origin at $t = 0$ s.

Die figuur toon die snelheidsgrafiek van 'n trein wat begin by die oorsprong by $t = 0$ s.



- (a) Find the acceleration of the train at $t = 3.0$ s.
Vind die versnelling van die trein by $t = 3.0$ s.

(2)

2

$$\text{Acceleration} = \text{Gradient} = a_{\text{avg}}$$

$$a_{\text{avg}} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{0 \text{ m/s} - (-2 \text{ m/s})}{4 \text{ s} - 0 \text{ s}}$$

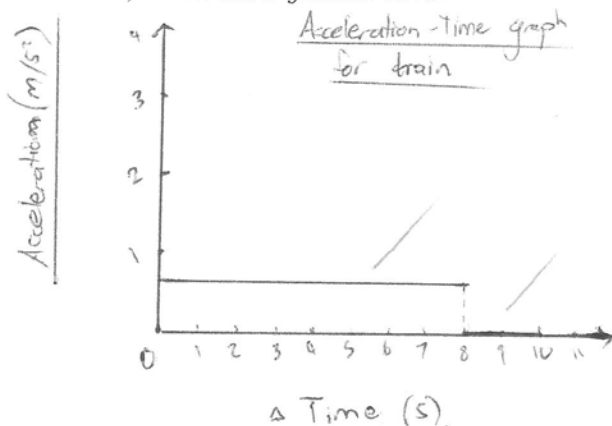
$$= 0,5 \text{ m/s}^2$$

- (b) Draw the corresponding position and acceleration graphs for the train. Your graphs must be neat, and to scale.

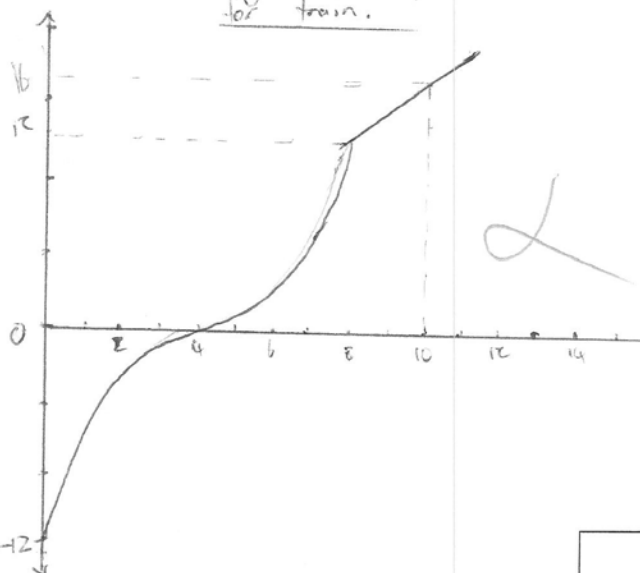
Teken die ooreenstemmende posisie- en versnellingsgrafieke vir die trein. Jou grafieke moet netjies wees, en tot skaal geteken word.

(6)

2



Displacement-Time Graph for train.



$$x_f = x_i + v_{xi}t + \frac{1}{2}at^2$$

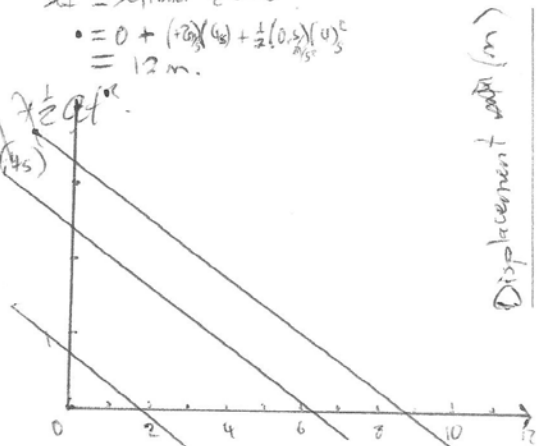
$$= 0 + (0 \text{ m/s})(4 \text{ s}) + \frac{1}{2}(0,5 \text{ m/s}^2)(4 \text{ s})^2$$

$$= 12 \text{ m}$$

$$x_f = x_i + v_{xi}t + \frac{1}{2}at^2$$

$$= 0 + 0 + \frac{1}{2}(0,5 \text{ m/s}^2)(4 \text{ s})^2$$

$$= 12 \text{ m}$$



4 Question 4 / Vraag 4

[5]

5

A particle is confined to move along the x -axis of a coordinate system with **constant** acceleration. If the position of the particle is given by $x = x_i + v_{xi}t + \frac{1}{2}a_x t^2$ at any time t , and the velocity of the particle is given as $v_x = v_{xi} + a_x t$ for any time t , prove that $v_x^2 = v_{xi}^2 + 2a_x \Delta x$. You must use the two given equations in your derivation.

'n Deeltjie is beperk om te beweeg langs die x -as van 'n koördinaat-stelsel met **konstante** versnelling. Indien die posisie van die deeltjie gegee word deur $x = x_i + v_{xi}t + \frac{1}{2}a_x t^2$ vir enige tyd t , en die snelheid van die deeltjie word gegee deur $v_x = v_{xi} + a_x t$ vir enige tyd t , bewys dat vir enige tyd t is $v_x^2 = v_{xi}^2 + 2a_x \Delta x$. Jy moet die twee gegewe vergelykings in jou afleiding gebruik.

$$x = x_i + v_{xi}t + \frac{1}{2}a_x t^2 \quad (1)$$

$$v_x = v_{xi} + a_x t \quad (2)$$

$$\text{Using (2): } t = \frac{v_{xf} - v_{xi}}{a_x} \quad (3) \quad \checkmark$$

$$\text{and Using (2): } a_x = \frac{v_{xf} - v_{xi}}{t} \quad (4)$$

→ sub (4) into (1):

$$x = x_i + v_{xi}t + \frac{1}{2}a_x t^2$$

$$x = x_i + v_{xi}t + \frac{1}{2}\left(\frac{v_{xf} - v_{xi}}{t}\right)t^2$$

$$x = x_i + v_{xi}t + \frac{1}{2}(v_{xf} - v_{xi})t \quad (5)$$

Then subs (3) into (5):

$$x = x_i + v_{xi}\left(\frac{v_{xf} - v_{xi}}{a_x}\right) + \frac{1}{2}(v_{xf} - v_{xi})\left(\frac{v_{xf} - v_{xi}}{a_x}\right)$$

$$x = x_i + \frac{v_{xf}v_{xi} - v_{xi}^2}{a_x} + \frac{1}{2a_x}(v_{xf}^2 - v_{xf}v_{xi} + v_{xf}v_{xi} - v_{xi}^2)$$

$$v_{xf}^2 = v_{xi}^2 + 2a_x(x_f - x_i)$$

$$v_{xf}^2 = v_{xi}^2 + 2a_x \Delta x$$

5 Question 5 / Vraag 5

[6]

A lead ball is dropped into a lake from a diving board 5.0 m above the water. After entering the water, it sinks to the bottom with a constant velocity equal to the velocity with which it hit the water. The ball reaches the bottom 3.0 s after it is released. How deep is the lake? Be sure to make a sketch, and show clearly your choice of coordinate system.

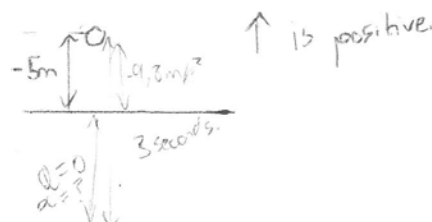
A lood-bal word in 'n meer laat val vanaf 'n duikplank wat 5.0 m bo die water is. Nadat die bal die water tref, sink dit na die bodem van die meer met 'n konstante snelheid gelyk aan die snelheid waarmee dit die water getref het. Die bal tref die bodem 3.0 s nadat dit losgelaat is. Hoe diep is die meer? Teken 'n skets, en dui duidelik jou keuse van koördinaatstelsel aan.

$$t = 3 \text{ s}$$

$$v_i = 0 \text{ m/s}$$

$$\Delta x = -5 \text{ m}$$

$$a = -9.8 \text{ m/s}^2$$



$$x_f = x_i + \frac{1}{2}(v_{xi} + v_{xf})t$$

$$v_{xf}^2 = v_{xi}^2 + 2a_x(x_f - x_i)$$

$$-5 \text{ m} = 0 \text{ m} + \frac{1}{2}(0 \text{ m/s} + v_{xf})t$$

$$v_{xf}^2 = 0 \text{ m/s}^2 + 2(-9.8 \text{ m/s}^2)(-5 - 0)$$

$$-5 \text{ m} = \frac{1}{2}v_{xf}t$$

$$v_{xf} = -9.9 \text{ m/s} \text{ as it hits the water.}$$

$$x_f = x_i + \frac{1}{2}(v_{xi} + v_{xf})t.$$

$$-5\text{ m} = 0 + \frac{1}{2}(0\text{ m/s} + (-9,9\text{ m/s}))t.$$

$$t = 1,01\text{ seconds}$$

$$\therefore 3\text{ seconds} - 1,01\text{ seconds} = 1,99\text{ seconds in water.}$$

$$\therefore t \text{ in water} = 1,99\text{ seconds.}$$

At constant velocity:

$$x_f = x_i + v_x t.$$

$$x_f = 0\text{ m} + (-9,9\text{ m/s})(1,99\text{ seconds})$$

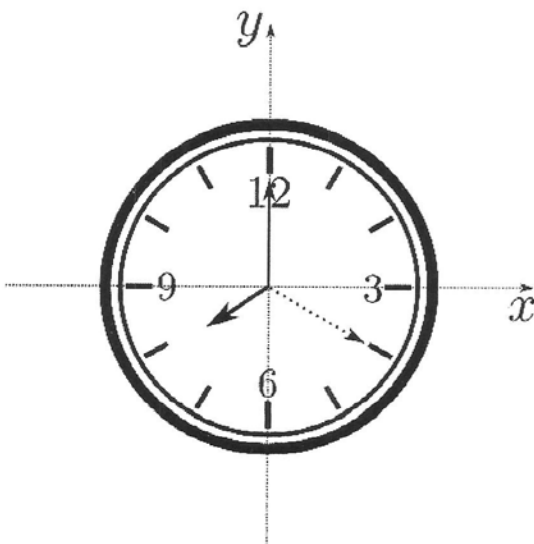
$$x_f = -19,701\text{ m.}$$

$\therefore$ The lake is 19,70 m deep.

6 Question 6 / Vraag 6

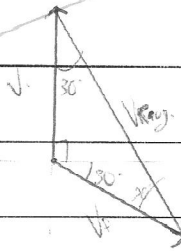
[9]

The minute hand on an analog watch is 2.0 cm in length. What is the **average** velocity vector, $\vec{v}_{\text{avg}}$, of the tip of the minute hand if it moves from 8:00 to 8:20? Determine the magnitude and direction (with respect to the positive x-axis) for this average velocity vector. (The dashed minute hand is shown for clarity in the figure). What is the average velocity vector if the minute hand moves from 8:00 to 9:00? Die minuutwyser op 'n analoog-horlosie is 2.0 cm lank. Wat is die **gemiddelde** snelheidsvektor, $\vec{v}_{\text{avg}}$, van die punt van die minuutwyser as dit beweeg vanaf 8:00 na 8:20? Bepaal die grootte en rigting (ten op sigte van die positiewe x-as) vir hierdie gemiddelde snelheidsvektor. (Die gestippelde minuutwyser word in die figuur getoon vir duidelikheid). Wat is die gemiddelde snelheidsvektor indien die minuutwyser vanaf 8:00 na 9:00 beweeg?



$$\vec{V}_{avg} = \frac{\Delta x}{\Delta t}$$

$$\vec{V}_{avg} = \frac{x_f - x_i}{t_f - t_i}$$



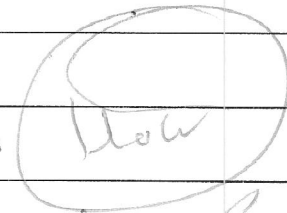
By means of sketch:

$$\Delta x = 3,5 \text{ cm}$$

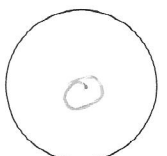
$$\vec{V}_{avg} = \frac{0,035}{20 \text{ min} \times 10^{-3} \text{ s}}$$

$$= 2,92 \times 10^{-5} \text{ m/s}$$

300° anti-clockwise from the positive x-axis

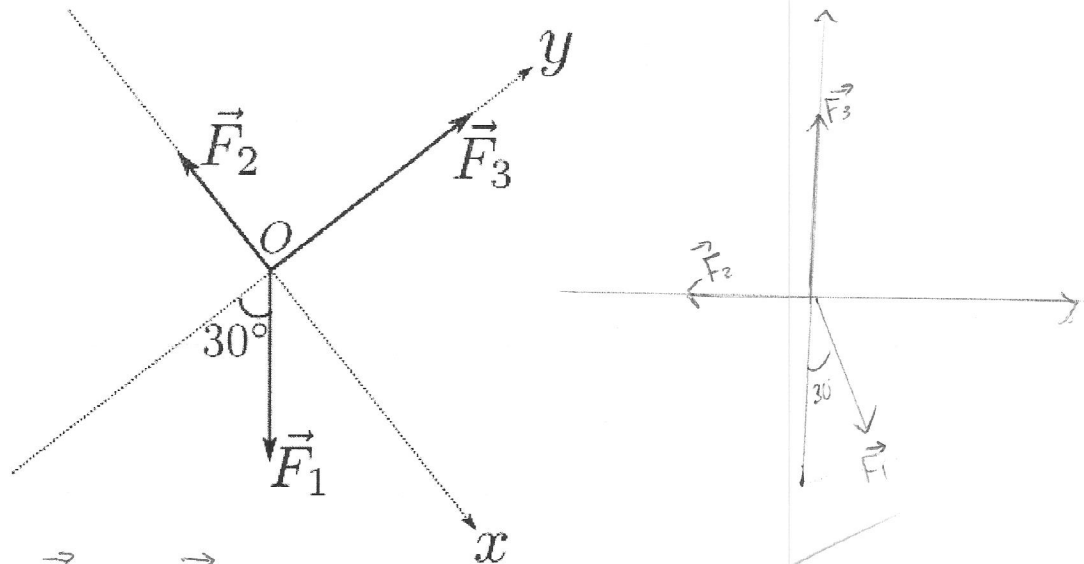


The average velocity of the minute hand moves from 08:00 to 09:00 is 0. because $\vec{V}_{avg} = \frac{x_f - x_i}{t_f - t_i} = \frac{12 - 12}{60/60 - 0} = 0$



Three vectors are shown in the figure, but they are not drawn to scale. The sum of the three vector is $\vec{R} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$. If $R_y = 0$ and $F_2 = 0.5 F_3$, find R_x in terms of F_1 .

Drie vektore word in die figuur getoon, maar is nie volgens skaal geteken nie. Die som van die drie vektore is $\vec{R} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$. Indien $R_y = 0$ en $F_2 = 0.5 F_3$, vind R_x in terme van F_1 .



$$\vec{R} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$$

$$R_y = 0 \quad \text{and} \quad R_{\text{sum}} = R \sin \theta$$

$$F_2 = 0.5$$

$$R_y = 0$$

$$\therefore F_1 \cos 30^\circ = F_3$$

$$\vec{R}_y = F_3$$

$$\vec{R}_x = R_y =$$

$$\therefore \vec{R}_x = \vec{F}_2 + (-F_1 \sin 30^\circ)$$

$$R_x = 0.5 - F_1 \sin 30^\circ$$

$$\text{But } \vec{R} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$$

$$\therefore \vec{R} = \vec{F}_1 + 0.5 + F_1 \cos 30^\circ$$

8

Question 8 / Vraag 8

[7]

If the vector components of a particle's position moving in the xy plane as a function of time are $\vec{x} = bt^2\hat{i}$ and $\vec{y} = ct^3\hat{j}$, where b and c are positive constants with appropriate dimensions such that the components will be in meters, at what time t is the angle between the particle's velocity and the x -axis equal to 45° ?

'n Deeltjie wat in die xy -vlak beweeg, se vektor-komponente van die posisie word as 'n funksie van die tyd gegee as $\vec{x} = bt^2\hat{i}$ en $\vec{y} = ct^3\hat{j}$. b en c is positiewe konstantes met die korrekte dimensies sodat die komponente in meters gemeet kan word. By watter tyd t is die hoek tussen die deeltjie se snelheid en die x -as gelyk aan 45° ?

$$\vec{x} = bt^2 \hat{i}$$

$$\vec{y} = ct^3 \hat{j}$$

$$\vec{x} = \vec{y}$$

at 45° to positive x -axis.

$$\therefore bt^2 \hat{i} = ct^3 \hat{j}$$

$$b = ct$$

$$b = ct$$

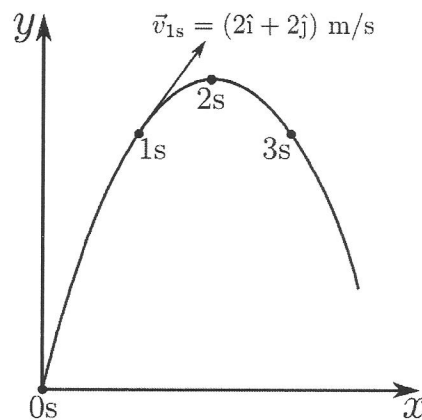
$$t = \frac{b}{c} \text{ seconds}$$

9 Question 9 / Vraag 9

[14]

A physics student on Planet Exidor throws a ball, and it follows the parabolic path as shown in the figure. The ball's position is shown at 1 s intervals until $t = 3$ s. At $t = 1$ s, the ball's velocity is $\vec{v}_{1s} = (2\hat{i} + 2\hat{j})$ m/s.

'n Fisika-student op Planeet Exidor gooi 'n bal, en dit volg die paraboliese pad soos aangetoon in die figuur. Die bal se posisie word gewys vir 1 s intervalle tot by $t = 3$ s. By $t = 1$ s, is die bal se snelheid $\vec{v}_{1s} = (2\hat{i} + 2\hat{j})$ m/s.



- (a) Determine the ball's velocity, in vector notation, at $t = 2$ s and $t = 3$ s.
Bepaal die bal se snelheid, in vektor notasie, by $t = 2$ s en $t = 3$ s.

(4)

~~$\vec{v}_{2s} = (2\hat{i} + 0\hat{j})$~~

$$\vec{v}_{2s} = (2\hat{i} + 0\hat{j})$$

$$\vec{v}_{3s} = (2\hat{i} - 2\hat{j})$$

- (b) What is the value of g on Planet Exidor?
 Wat is die waarde van g op Planeet Exidor?

* up is positive.

(4)

0

$$V_{yf} = V_{yi} + a_y t.$$

$$-2\hat{j} = 0\hat{j} + a_y(1).$$

$$a_y = -2\hat{j}$$

The value of g on Planet Exidor

$$2\hat{j}/s^2$$

- (c) Determine the ball's velocity at $t = 0$ s.
 Bepaal die bal se snelheid by $t = 0$ s.

(3)

2 1/2

$$V_{yf} = V_{yi} + a_y t.$$

$$0\hat{j}/s = V_{yi} + (-2\hat{j}/s^2)(2).$$

$$V_{yi} = 4\hat{j}/s \text{ upwards}$$

$$\therefore V_i = (2\hat{i} + 4\hat{j})$$

- (d) Determine the ball's launch angle.
 Bepaal die bal se lanseer-hoek.

(3)

3

$$\tan \theta = \frac{4\hat{j}}{2\hat{i}}$$

$$\theta = 63.43^\circ \text{ above positive } x\text{-axis.}$$

END OF TEST / EINDE VAN TOETS

5 1/2